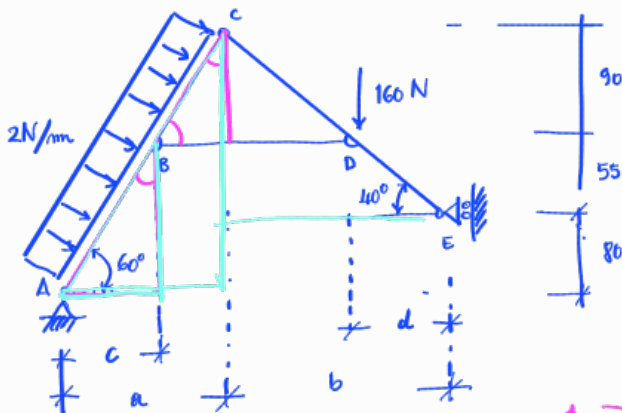




$$\overline{AB} = \frac{80+55}{\sin 60^\circ} \quad \left. \begin{array}{l} \overline{BC} = \frac{90}{\sin 60^\circ} \end{array} \right\} \overline{AC} = \overline{AB} + \overline{BC} = 259.8 \text{ mm}$$

$$\overline{CD} = \frac{90}{\sin 40^\circ} \quad \left. \begin{array}{l} \overline{DE} = \frac{55}{\sin 40^\circ} \end{array} \right\} \overline{CE} = \overline{CD} + \overline{DE} = 225.6 \text{ mm}$$

$$a = \frac{90+55+80}{\tan 60^\circ} \quad c = \frac{55+80}{\tan 60^\circ}$$

$$b = \frac{90+55}{\tan 40^\circ} \quad d = \frac{55}{\tan 40^\circ}$$

Eq. cardinali delle strutture:

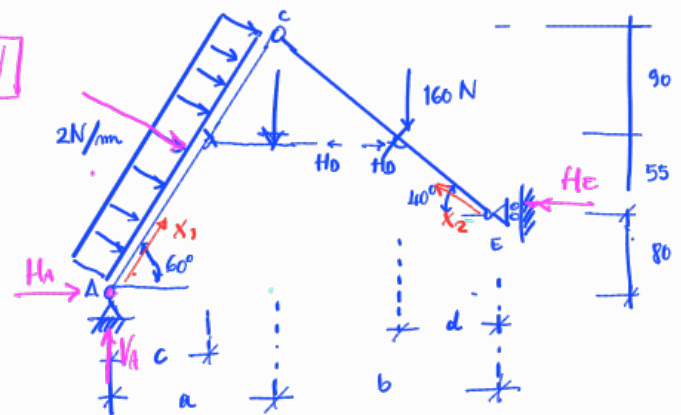
$$\Sigma H = 0 \quad H_A - H_E + 2 \cdot \overline{AC} \cdot \sin 60^\circ = 0 \rightarrow H_A = 867.9 \text{ N}$$

$$\Sigma V = 0 \quad V_A - 160 - 2 \cdot \overline{AC} \cdot \cos 60^\circ = 0$$

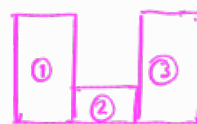
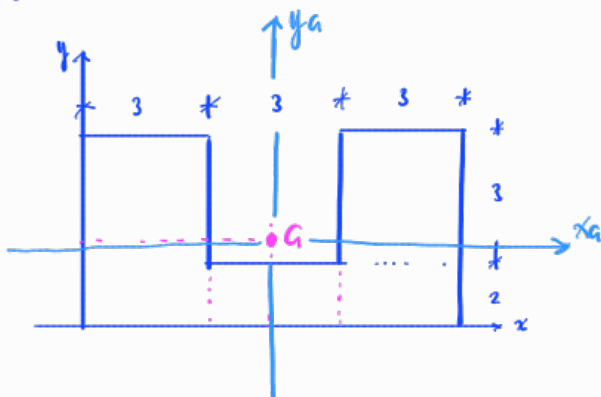
$$\rightarrow V_A = +160 + 2 \cdot \overline{AC} \cdot \cos 60^\circ = 419.8 \text{ N}$$

$$\Sigma M_A = 0 \quad + 2 \cdot \frac{\overline{AC}^2}{2} - H_E \cdot 70 + 160(a+b-d) = 0$$

$$\rightarrow H_E = 1317.9 \text{ N}$$



Geometria delle masse



$$A_{TOT} = 36a^2$$

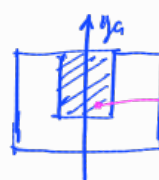
$$S_x = A_1 \cdot y_{G1} + A_2 \cdot y_{G2} + A_3 \cdot y_{G3} = (15a^2 \cdot \frac{5}{2}a) + (a^2 \cdot a) + (15a^2 \cdot \frac{5}{2}a) = 81a^3$$

$$S_y = A_1 x_{G1} + A_2 x_{G2} + A_3 x_{G3} = (15a^2 \cdot \frac{3}{2}a) + (a^2 \cdot \frac{9}{2}a) + (15a^2 \cdot \frac{15}{2}a) = 162a^3$$

$$x_G = \frac{S_y}{A_{TOT}} = \frac{162a^3}{36a^2} = \frac{9}{2}a$$

$$y_G = \frac{S_x}{A_{TOT}} = \frac{81a^3}{36a^2} = \frac{9}{4}a$$

$$\begin{array}{l|l|l} x_{G1} = \frac{3}{2}a & x_{G2} = \frac{9}{2}a & x_{G3} = \frac{15}{2}a \\ y_{G1} = \frac{5}{2}a & y_{G2} = a & y_{G3} = \frac{5}{2}a \\ A_1 = 15a^2 & A_2 = a^2 & A_3 = 15a^2 \end{array}$$

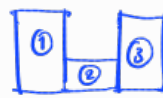


$$J_{y_{G1}} = \frac{b h^3}{12} = \frac{5(9)^3}{12} a^4 = \frac{3645}{12} a^4$$

$$J_{y_{G2}} = \frac{b h^3}{12} = \frac{3 \cdot (3)^3}{12} a^4 = \frac{81}{12} a^4$$

$$J_{y_G} = J_{y_{G1}} - J_{y_{G2}} = \left(\frac{3645 - 81}{12} \right) a^4 = 297 a^4$$

$$J_{x_0} = J_x - A y_c^2$$



$$J_x = J_{x_1} + J_{x_2} + J_{x_3}$$

$$\begin{cases} J_{x_1} = \frac{bh^3}{3} = \frac{3(5)^3}{3} a^4 = \frac{375}{3} a^4 \\ J_{x_2} = \frac{bh^3}{3} = \frac{3(2^3)}{3} a^4 = \frac{24}{3} a^4 \\ J_{x_3} = J_{x_1} = \frac{375}{3} a^4 \end{cases}$$

$$J_x = \frac{724}{3} a^4$$